

Running a successful farming business in the arid regions of Queensland is tough and maximising the productivity of your land is essential. Nobody knows this like rural business owners Nichole James and Jake & Adam Fietz.

Nichole, Jake and Adam are participating in a field trial being delivered by Central Highlands Regional Resources Use Planning Cooperative (CHRRUP) as part of the Future Drought Fund (FDF) Drought Resilient Soils and Landscapes program. The project aims to demonstrate drought resilient grazing practices that can rehydrate grazing landscapes and improve soils and water on farms. This supports building drought resilience by enhancing total biomass, reducing erosion and contributing to increased productivity of grazing enterprises.

Nichole's property is one of five included in the project. Glenn Landsberg, a Regeneration Consultant, has been working with Nichole to modify her farm landscape for better water retention, particularly focusing on the restoration of gullies.

'Recovery here isn't really going to happen until we intervene and make something happen' Glenn said.

After working with Glenn and employing earthworks to retain water across her property, Nichole has also implemented smaller fenced paddocks and rotational grazing to allow soil and pasture to rest and recover.

'Now we've done earthworks, it'll help make it recover quicker and slow the erosion. Now that I've split the paddock, I can control the grazing pressure even more. With these earthworks, which will also help with slowing the water, we should see a much bigger recovery in a much shorter time' says Nichole.

At 'Hiddenvale' in Queensland, Jake and Adam's property is also part of the project and already is seeing results through effective water management, which reduces soil erosion and surface run-off.

'The drought over the last 10 years has just knocked us about. It's just mind blowing how much it has actually knocked us around, just so much destruction. I don't want to ever be in a nasty situation again where we have to destock. If we have to destock again, we'd have to sell the place basically' Jake Fietz said.

Adam adds, 'This was all deeply eroded, lots of dead trees and a lot of water run-off. We used the dead trees and wind-rowed them up, slowed the water run-off. That caught some seed, and this is what's happened. I've got them about 80 metres apart all the way up. And well... you can see....it's all grass now. It stopped a lot of the erosion which is always a good thing'.

See how the Future Drought Fund and CHRRUP are providing support to help Central Queensland farmers restore natural ecosystems to better manage water and build drought resilience

Video duration: 10mins 16 secs

Introduction

This is the transcript of a video case study produced by the Central Highlands Regional Resources Use Planning Cooperative (CHRRUP) for a program funded by the Australian Government's Future

Drought Fund (FDF), the Drought Resilient Soils and Landscapes program. It outlines how via support from the FDF, farmers in Central Queensland are working with a Regeneration Specialist to restore gullies, better manage water, reduce erosion and heal country.

Learn more about the [Drought Resilient Soils and Landscapes Grants Program](#).

[Recording begins]

Voiceover [00:16]:

Building drought resilience in the arid regions of Queensland, relies on a holistic approach that considers many aspects of water, soil, pasture and cattle management. This video describes approaches to managing soil erosion and water conservation.

Nichole James [00:17]:

So we have the rain gauges that are roughly 8 kilometres square, and you can have one shower at the house, and I had 16 mil at the house, then 2 mil up on the ridge, then I might have 32 over in the bull paddock, then there might be nothing down at the Dry Alice. So, it's half handy to know what sort of rain I've actually had out in the paddocks, not just what was at the house.

So, this is the Avenza map, and it shows a bit of item off our boundary, so with this gully here that we're going to be playing with for the rehydration project, if you get a heavy enough shower up there, this gully can run quite fast. So, yeah you get a bit of erosion and what not. And then the country that leads off from the gully is sort of the sweetest bit in the paddock, so the cows tend to like to sit on here, so it tends to get flogged a bit, and has turned into a bit of clay pan. And with rotational grazing, it has recovered a bit over the years.

Glenn Landsberg [01:24]:

All this soil has been lost from this area here, and this is now the level of the water table in the soil. So, a lot of trees up higher on the flat either side no longer have their roots in soil moisture, and we can see the likes of this tree here, that in it's lifetime, this is where the soil used to be up here... which corresponds with the flat level soil across here. There are a lot of these exposed trees and exposed roots all the way down here, but the issue now is all of the moisture in the flats on either side, the soil moisture or the water table is draining to the level of the erosion at the bottom here, which is why so many of the trees either side are dying up on the flats. Recovery here isn't really going to happen, until we intervene and make something happen.

So, we're looking here at where we've got the central erosion feature in the middle, and we've got all these other little fingers... because the water's escaping so quickly, getting sucked out of this area so quickly, out of that main erosion, that it's now sending out all these fingers up into the other flats up on the sides of the hills and all that extra water, and it's all because that waters escaping so quickly down that central feature.

This sort of snowballing of self-destruction, because as the plants die out along the edges those couple of metres either side of this that are out of moisture, all this bare ground will start to erode, it will all start to fall into this.

Nichole James [02.47]:

When we do a bit of earthworks here in the next few days, it'll help make it recover quicker and slow the erosion up. And now that I've split the paddock up I can control the grazing pressure on it even more, and now with these earthworks we are doing, which will also help with slowing the water up and keeping it on here, we should see a much bigger recovery in a much shorter time.

Glenn Landsberg [03.16]:

What Nichole was dealing with initially with very large paddocks, they weren't able to have rotation and rest and recovery. But now that she has the smaller paddocks, having the wire investment and the water investment, she can rotate spell country. The increase in pasture productivity, for having that rest and recovery, is really noticeable driving onto place.

We are just putting a wall across here, and putting a bank across it, and then we are running spreading channels out. We've got some really good country either side that's well grassed and well established and we're going to run spreader channels out on the contour, so no fall once we get out. When those spreader channels are full of water, the deeper we can get them the better, within reason. It always depends what machinery you've got available, but we want to be able to carry sufficient volume to the end of our spreader channels. Once those channels are full, because they are on the contour, they're just going to evenly overflow and water the grass. We've taken the energy out of the water and then we're spreading it out safely and getting it back onto the pasture.

So why are we doing it? We can certainly vastly improve the productivity of the pasture that we are going to put that extra water onto. This bottom section here was starting to heal itself to some degree, but most of what's above us, is only going to continue to get worse.

This is a little area here where we don't need to do big interventions, but for all these little scalded areas, they all start running water and when they accumulate next thing you know, we've got gully's. So, we are getting in early here, we are going to pre-empt the gully stage. And on all these little bald patches we are doing to do a bit of a ripping program. So, this is the ripping process here, it's simply putting the rippers in and giving an entry point for the water to get into the soil. It will create enough of a little disturbance, it will act like little mini contours and the surface runoff is going to hit those little contours and start to get into the soil for us. So, we are doing them on the contour, very important to do them on the contour.

But as that's built up into a big pool of dead water, once it gets full it's got to go somewhere and this is where it's going to go. This is where the overflow is. The water is going to gently start coming out here and it's going to flow through, in this case we've made a v-drain, that is all on the contour. So, that is the opening coming through, the water is going to run down into here, it's going to fill that v-drain the whole way along it, and then once the v-drain is full then it will start to gently overflow on the bottom side and just start watering all that grass.

So, we've taken an overflow, like that whole little eroded flat might of been 70 metres wide and all of the water had travelled down that 70 metres. Now we've got 700 metres out of this side of overflow, a similar amount on the other side. All the water is still going to be able to travel down, but it's spread over such an area and so gently through all that grass that we've taken all the damage and impact out of it.

I've gone around so many properties and you get the opportunity to see what's worked and what hasn't for other people. I haven't had to make all the mistakes and I've been able to learn from other people's wins. Particularly of importance is the machinery operators, they know the soil types, they know the different areas so I'm constantly learning every time I go somewhere talking to the people driving the machinery. What the soil structure is, how it's going to respond, what we've got to be careful of. Here we struggled a little bit, it's very sandy up further, and the grader, the sand was spilling out around the front of blade on it, so we had to modify our approach to manage that sand, and how hard it was to push it up where we needed it.

So, what we're seeing here is a structure that is suitable for this landscape. This spreader channel here runs three quarters of a kilometre. If we had fall in it, if we didn't have our levels done properly, if we had a bit of a fall in the elevations or we had low spots in it, all the water would be funnelled to one spot and you'd just be transferring an erosion there and creating a new erosion feature out further. So, the channel you're seeing here is on the contour, it's perfectly level, it's laser levelled in, so there is no fall and we've eliminated the low spots as much as possible. If we didn't it wouldn't work, if it wasn't done with that level of accuracy. It's not something you can do off satellite imagery.

We can't just put these in and just think they are going to work on their own. Every landscapes going to be different, we've got to design things to suit a landscape and to suit the problems. But their only one element of it all, there's so many other things that make them work. There's got to be the management, the fencing and the monitoring the goes on as well. If little issues occur, it's just being quick to intervene before they get too bad, before any other major blowouts start to happen. And you've got to give them time to settle in and start to function well.

Jake Fietz [08.20]:

Yeah, so we've got [inaudible] here and I reckon we've lost about 75% of it. And just in the drought over the last 10 years, it's just knocked it about, it's not coming back to life. It's just mind blowing how much it has actually knocked us around. Yeah, just so much destruction.

So, I don't want to ever be in a nasty situation again where we have to destock and stuff. If we have to destock again, we'd have to sell the place basically.

Adam Fietz [08.56]:

It was only about 12 months ago, I was in the dozer pushing up all the dead timber into these windrows.

Glenn Landsberg [09.00]:

So, it was all bare soil?

Adam Fietz [09.03]:

Yeah, all scalded, burnt, hard as rock dirt. Well, it was just straight run-off. The water would flow, same as up the top there, the water would flow about this deep straight over and just taken about 50 mil off the dirt at a time and carry it elsewhere, probably most likely next door. There was actually nothing here basically.

Glenn Landsberg [09.26]:

You can actually see the silt starting to get deposited way back here isn't it? It's pretty much a silt drop from there all the way down to here. This was all deeply eroded.

Adam Fietz [09.36]:

Yeah, we'll go straight up here and duck across and have a look at the other side there. There are a lot of dead trees and a lot of water run-off. So, we thought we might as well use the dead trees and wind rowed them up, slow down the water, and hopefully catch some seed, and this what's happened.

I've got them about 80 metres apart all the way up. And where there was never any grass....it's all grass now. It stopped a lot of the erosion which is always a good thing'.

[Recording ends]

Ends screen displays wording that says this project is supported by CHRRUP, through funding from the Australian Government's Future Drought Fund and displays the logos of Department of Agriculture, Fisheries and Forestry and the Future Drought Fund and also the Central Highlands Regional Resources Use Planning Cooperative (CHRRUP).

Source: <https://www.agriculture.gov.au/about/news/saving-soil-water>